Supplementary Table 1: Correspondence between parameters named in the manuscript and variables in the associated code (publicly available on GitHub).

Description	Paper	Code
Recorded density	$D_{p,t}$	dens
SD of density at the log-scale	σ_{obs}	sigma_obs
Natural mortality rate	m	est_m
Fishing mortality rate	$f_{a,t}$	f
Intercept for detection probability	β_0	beta_obs_int
Coefficient linking detection probability to density	β_1	beta_obs
Overall average recruitment	μ	mean_recruits
Conditional variance in the AR1 process	σ_{proc}	sigma_r
White noise used in the AR1 process	z_t	raw
Autcorrelation from AR1 term	α	alpha
Excess in natural mortality due to temperature	γ	m_e
Isotropic dispersal rate	δ	d
Sea bottom temperature	$T_{p,t}$	sbt
How much tax per unit of temperature	β_{tax}	beta_t
Suitability index	$I_{p,t}$	T_adjust
Temperature optimizing $I_{p,t}$	τ	Topt
Width parameter from $I_{p,t}$	ω	width

Supplementary Table 2: Hyperparameters used in the DRM.

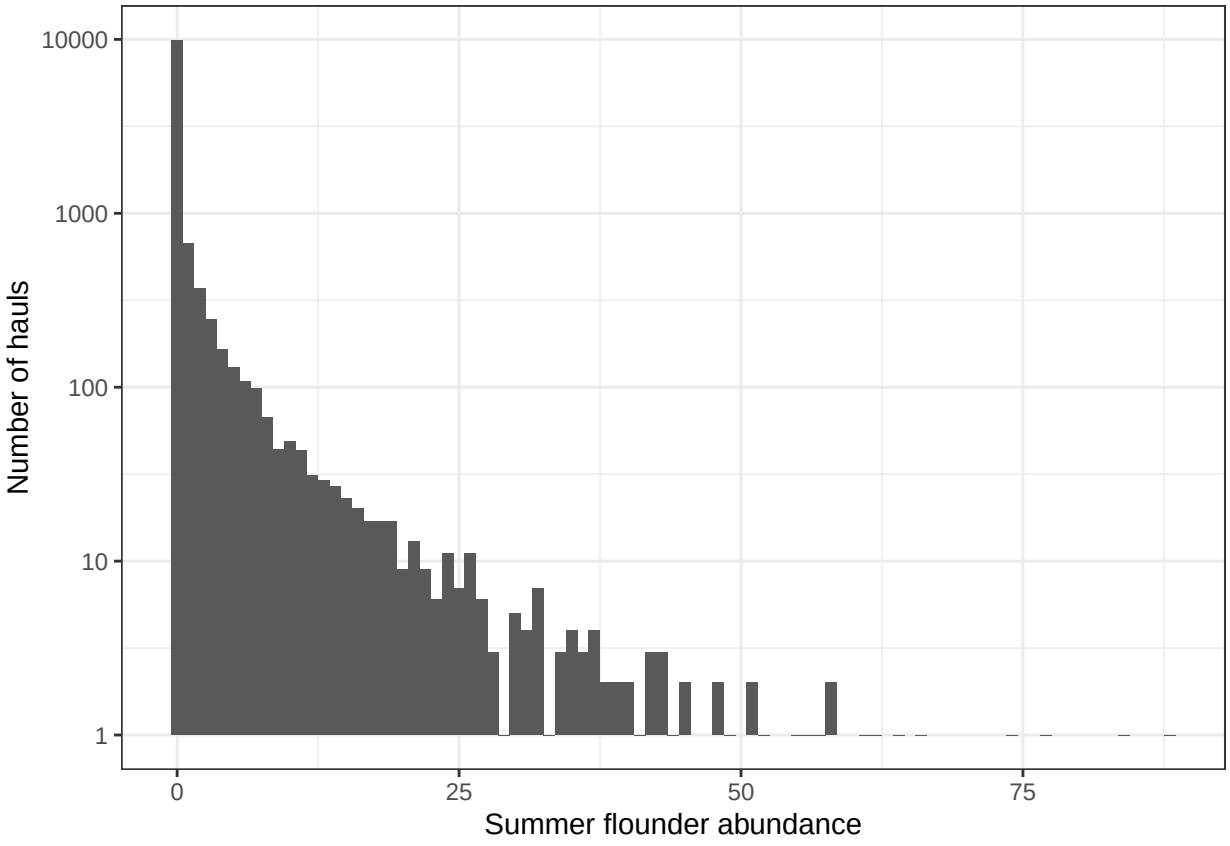
Parameter	Prior.Distribution
$\beta_{obs,1}$	Normal $\sim (0.001, 0.1)$
$\beta_{obs,0}$	Normal $\sim (-100, 4)$
z	Normal $\sim (0, 1)$
σ_{proc}	Normal $\sim (0.2, 0.1)$
σ_{obs}	Normal $\sim (0.2, 0.2)$
$width$	Normal $\sim (4, 2)$
T_{opt}	Normal $\sim (18, 2)$
d	Normal $\sim (0.01, 0.1)$
β_{tax}	Normal $\sim (0, 2)$
α	Beta $\sim (12, 20)$
μ_r	Lognormal $\sim (7, 5)$
m	Normal $\sim (0.25, 0.1)$
γ	Exponential ~ 2.3

Supplementary Table 3: Bayesian linear regressions of sea bottom temperature (measured in the trawl surveys) on year within each patch during the model training interval (1972-2006) with Gaussian errors, summarized by the median and 95% Highest Posterior Density Interval (HPDI) of the posterior estimate for the slope term. Values are truncated at three significant digits.

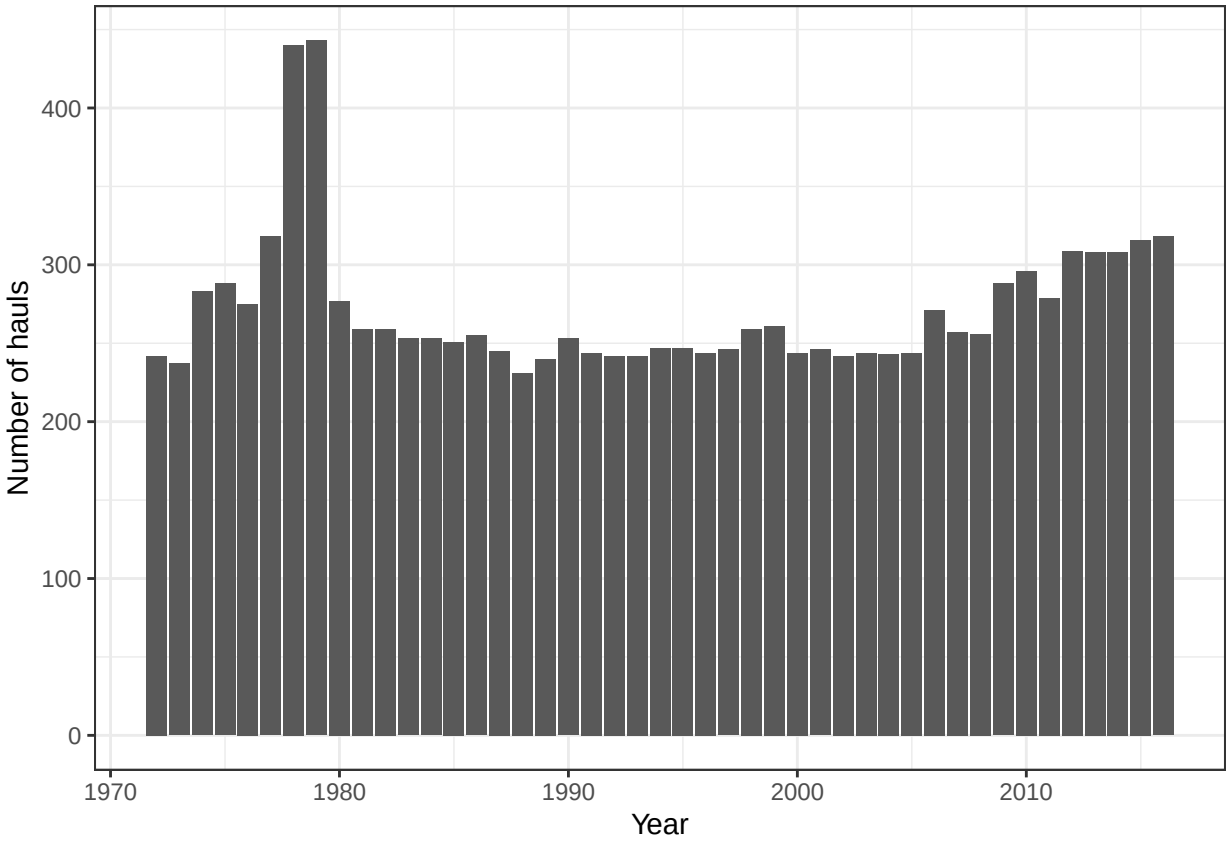
Patch	Median	HPDI
35	0.00236	-0.0734 — 0.0803
36	0.04540	0.00128 — 0.0909
37	0.05610	0.0106 — 0.0972
38	0.02900	-0.00582 — 0.0654
39	0.04710	0.0203 — 0.0709
40	0.01940	0.00837 — 0.0309
41	0.02680	0.0108 — 0.0428
42	-0.00264	-0.0103 — 0.00479
43	0.00328	-0.00566 — 0.0123
44	0.00607	-0.0154 — 0.0253

Supplementary Table 4: Bayesian linear regressions of sea bottom temperature (measured in the trawl surveys) on year within each patch during the model testing interval (2007-2016) with Gaussian errors, summarized by the median and 95% Highest Posterior Density Interval (HPDI) of the posterior estimate for the slope term. Values are truncated at three significant digits.

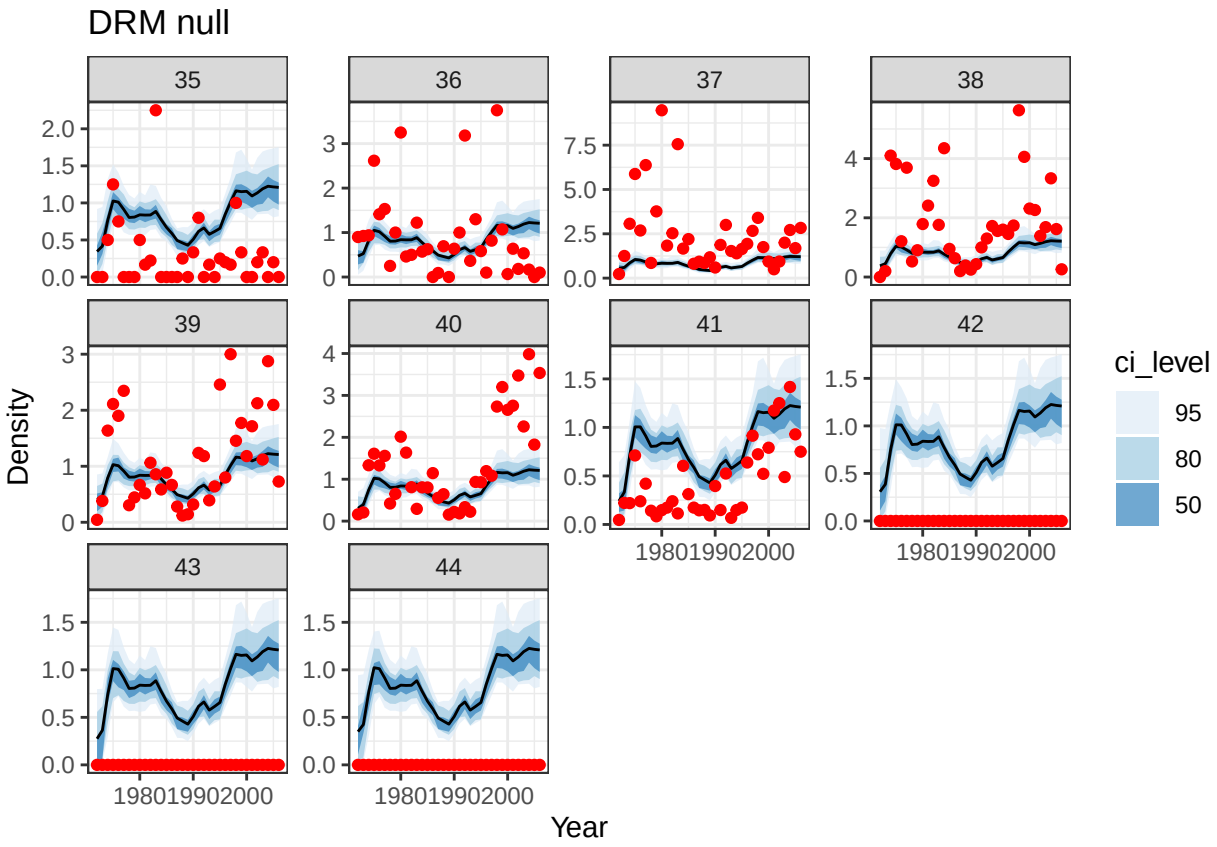
Patch	Median	HPDI
35	0.1520	-0.31 — 0.598
36	0.2310	-0.139 — 0.561
37	0.1620	-0.134 — 0.477
38	-0.0439	-0.266 — 0.17
39	0.0890	-0.0928 — 0.261
40	0.3610	0.276 — 0.446
41	0.1590	0.0441 — 0.278
42	0.1690	0.114 — 0.228
43	0.1740	0.106 — 0.243
44	0.2530	0.159 — 0.342



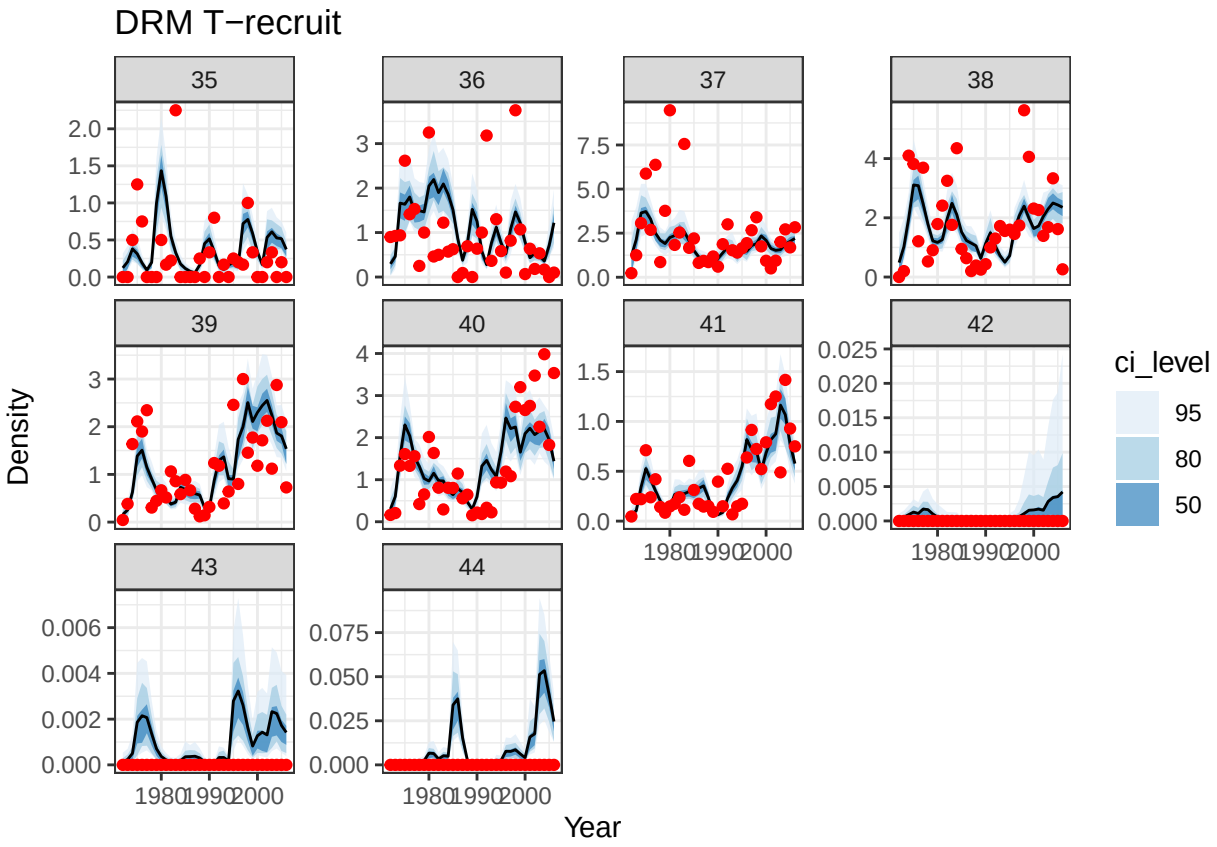
Supplementary Figure 1: Frequency distribution of summer flounder abundance in all hauls used in the analysis (testing and training data combined; n = 12,203).



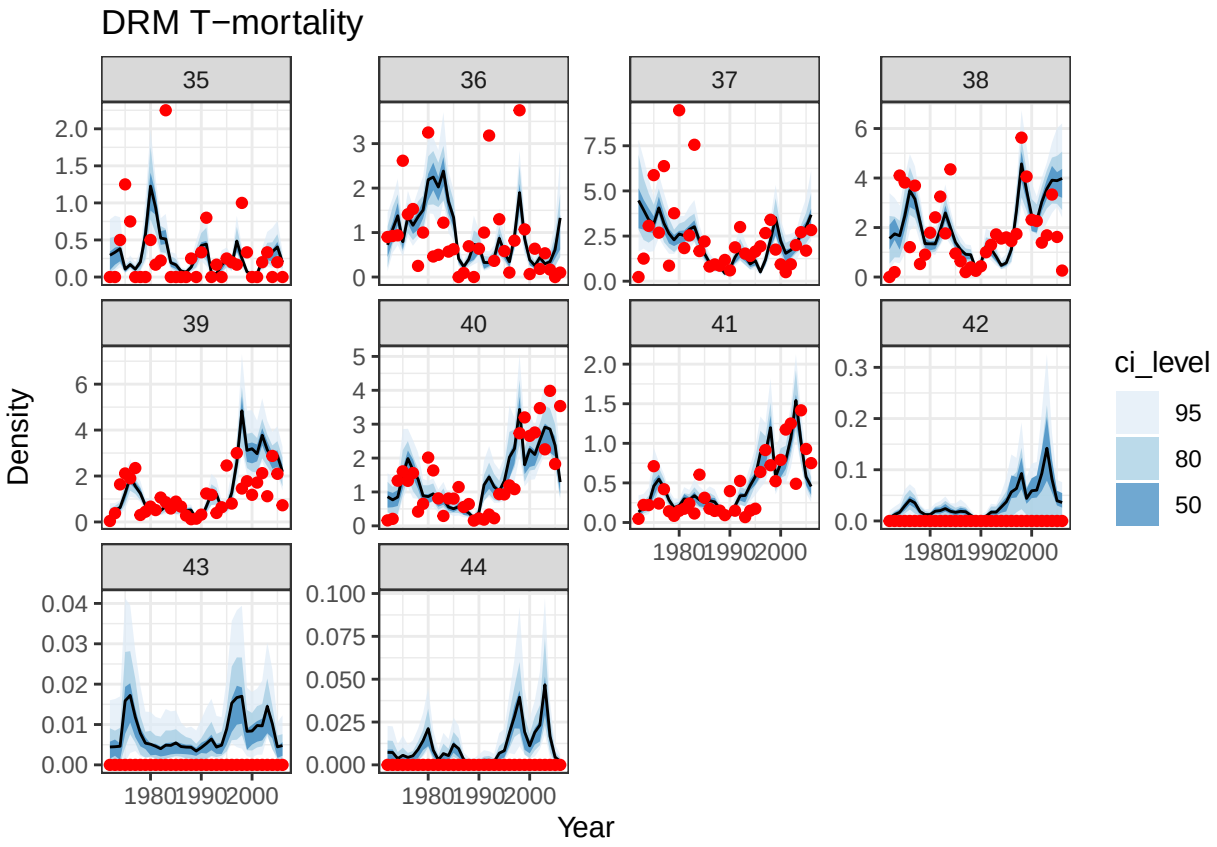
Supplementary Figure 2: Number of hauls per year; n = 12,203.



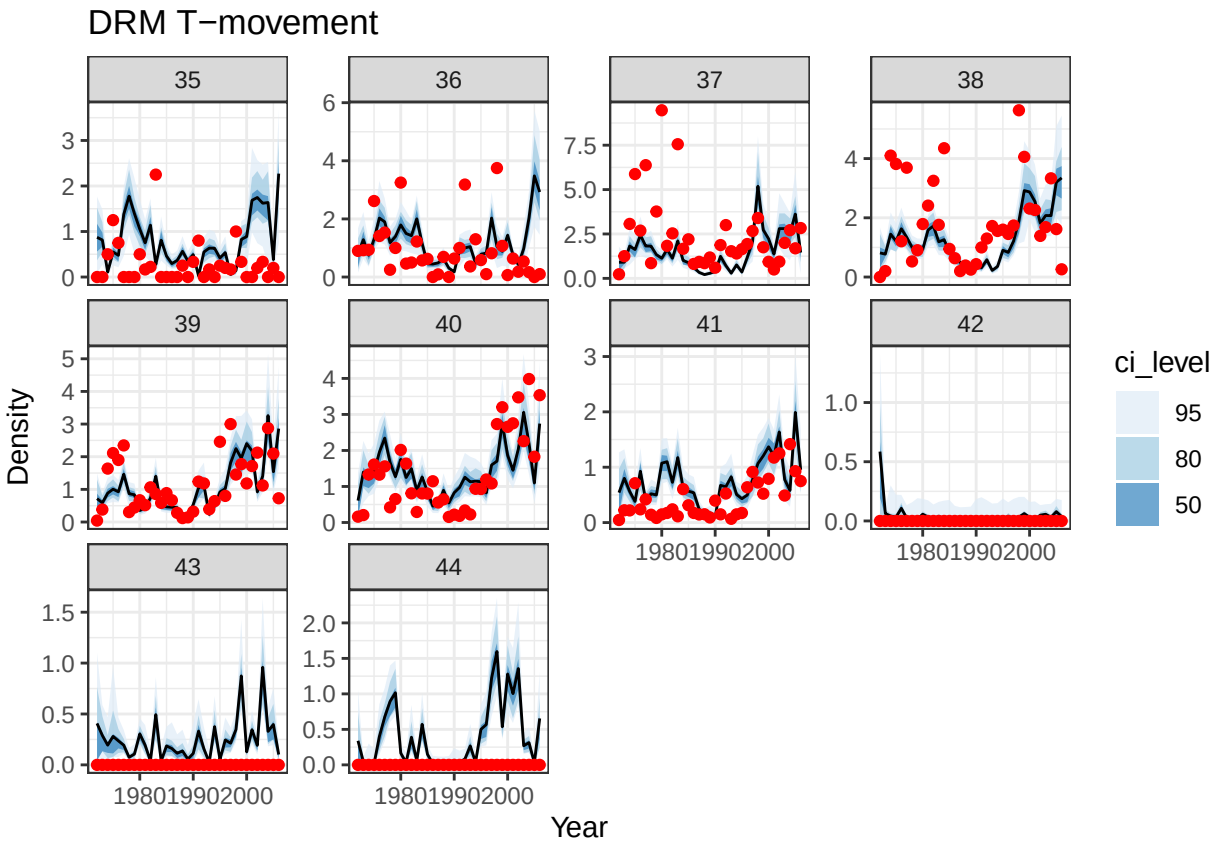
Supplementary Figure 3: Median density estimated by the null DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



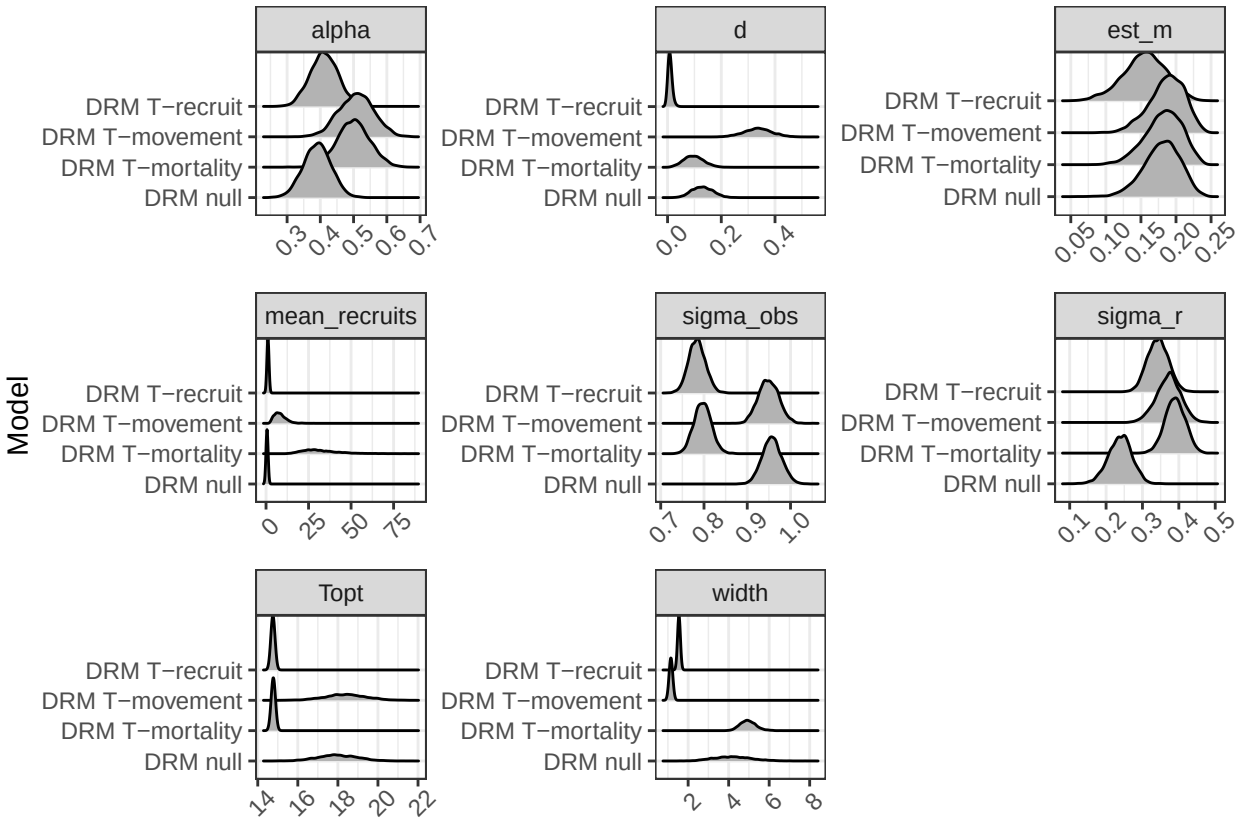
Supplementary Figure 4: Median density estimated by the temperature-dependent recruitment DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



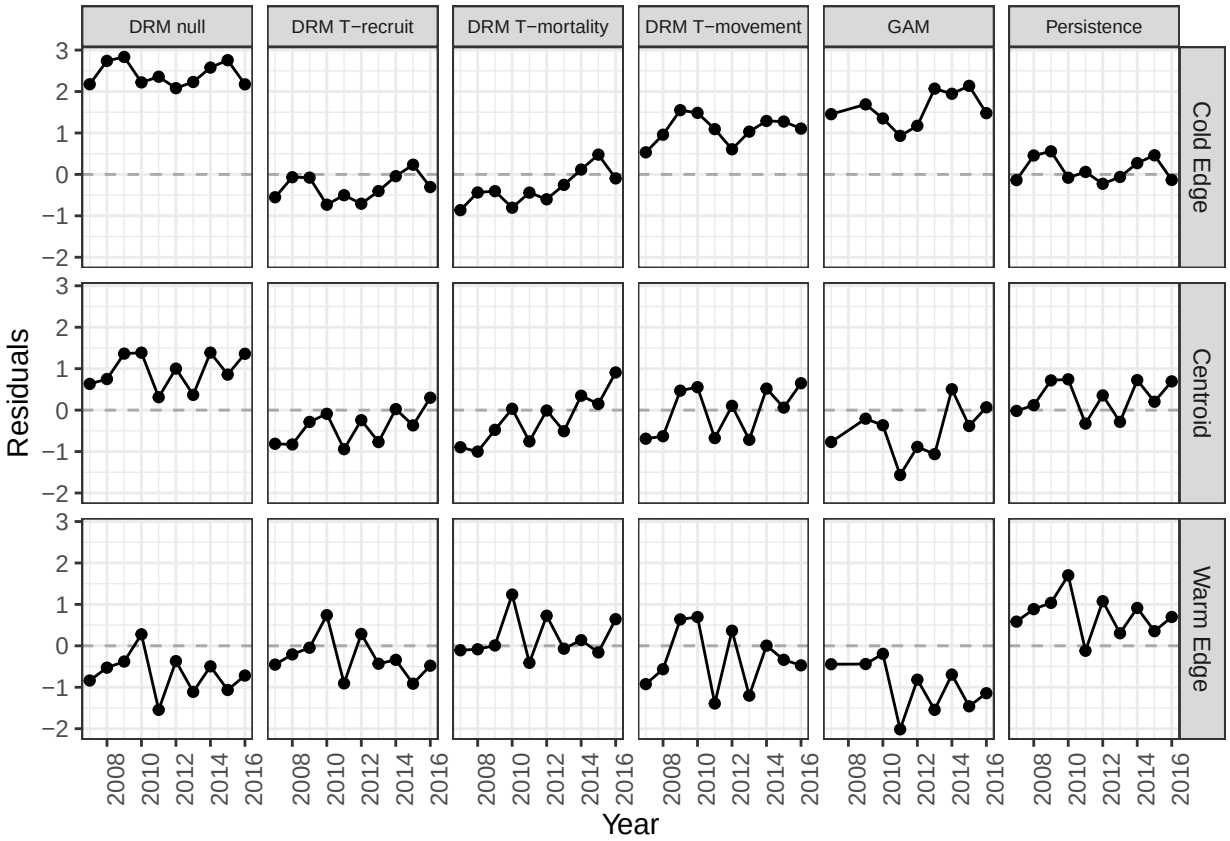
Supplementary Figure 5: Median density estimated by the temperature-dependent mortality DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



Supplementary Figure 6: Median density estimated by the temperature-dependent movement DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



Supplementary Figure 7: Posterior distributions of parameters from the four fitted DRMs. These parameters are d, the annual dispersal fraction between adjacent patches; mean_recruits, the average density of recruits per patch; sigma_obs, the standard deviation of density at the log scale; Topt, the optimal temperature estimated for whichever process (movement or recruitment or mortality) was estimated as temperature-dependent, in °C; and width, a parameter controlling how sensitive the modeled process was to temperature. See Supp. Tab. 1 for correspondence of code variables to model parameters in the manuscript.



Supplementary Figure 8: Residuals (estimated minus observed values) of range metrics each year over the ten-year testing period for the focal models in the main text: DRM with no temperature effect (DRM null) or a temperature effect on recruitment, mortality, or movement; a GAM SDM; and a persistence forecast. The horizontal dashed line indicates zero residuals, i.e., the estimate perfectly matches the testing data.

We used R v. 4.4.0 (R Core Team 2024a) and the following R packages: bookdown v. 0.42 (Xie 2016, 2025), brms v. 2.22.0 (Bürkner 2017, 2018, 2021), broom.mixed v. 0.2.9.7 (Bolker and Robinson 2026), cmdstanr v. 0.8.1 (Gabry et al. 2024), coda v. 0.19.4.1 (Plummer et al. 2006), data.table v. 1.16.4 (Barrett et al. 2024), directlabels v. 2024.1.21 (Hocking 2024), doParallel v. 1.0.17 (Corporation and Weston 2022), foreach v. 1.5.2 (Microsoft and Weston 2022), ggdist v. 3.3.2 (Kay 2024b, 2024a), ggh4x v. 0.3.0 (van den Brand 2024), ggrepel v. 0.9.6 (Slowikowski 2024), ggridges v. 0.5.6 (Wilke 2024), ggspatial v. 1.1.9 (Dunnington 2023), grid v. 4.4.0 (R Core Team 2024b), here v. 1.0.1 (Müller 2020), kableExtra v. 1.4.0 (Zhu 2024), knitr v. 1.49 (Xie 2014, 2015, 2024), Matrix v. 1.7.0 (Bates, Maechler, and Jagan 2024), mgcv v. 1.9.1 (S. N. Wood 2003, 2004, 2011; S. N. Wood et al. 2016; S. N. Wood 2017), parallel v. 4.4.0 (R Core Team 2024c), patchwork v. 1.3.0 (Pedersen 2024), posterior v. 1.6.0 (Vehtari et al. 2021; Bürkner et al. 2024), rmarkdown v. 2.29 (Xie, Allaire, and Grolemund 2018; Xie, Dervieux, and Riederer 2020; Allaire et al. 2024), rnaturalearth v. 1.0.1 (Massicotte and South 2023), rnaturalearthdata v. 1.0.0 (South, Michael, and Massicotte 2024), rstan v. 2.32.6 (Stan Development Team 2024), sf v. 1.0.19 (Pebesma 2018; Pebesma and Bivand 2023), tidybayes v. 3.0.7 (Kay 2024c), tidytext v. 0.4.2.9000 (Silge and Robinson 2016), tidyverse v. 2.0.0 (Wickham et al. 2019).

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